

- Fundamental Counting Principle
- permutations
- Combinations
- Probability

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The Fundamental Counting Principle

- If one event can occur in n_1 ways and a second event can occur in n_2 ways, the number of ways the two events can occur in sequence is $n_1 \times n_2$. This rule can be extended for any number of events occurring in sequence.

Example1:

- There are four blood types: A, B, AB and O. Also, blood can be Rh⁺ and Rh⁻. The donor can be classified as male and female. In how many ways can the donor be labeled
- $4 \times 2 \times 2 = 16$ ways

Example3:

- How many different patient ID are there that containing exactly two English letters and 3 numbers ?

1. each digit or letter can be used only once and not repeated?
2. each digit or letter can be repeated?

Solution

1. each digit or letter can be used only once and not repeated?

$26 \times 25 \times 10 \times 9 \times 8 = 468000$ possible.

2. Each digit or can be repeated.

$26 \times 26 \times 10 \times 10 \times 10 = 676000$ possible IDs.

An important application of the Fundamental Counting Principle:
determining the number of ways that n objects can be arranged in order (places=choices).

Which equals

$$n! = n \times (n-1) \times (n-2) \times \dots \times 1$$

- What about if we have n objects with only k places and $k \leq n$

Then number of ways

$$n \times (n-1) \times \dots \times (n-(k-1)) = \frac{n!}{(n-k)!}$$

The factorial

- The expression $n!$ is read as n factorial and is defined as follows:

$$n! = n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 3 \times 2 \times 1$$

Here are several values of $n!$.

As a special case, $0! = 1$

$$1! = 1$$

$$2! = 2 \times 1 = 2$$

$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Notice that as n increases, $n!$ becomes very large.

Example 4: Finding the number of ways to order of n objects in n places and k places

- In how many ways can 9 students set in 9 chairs

Solution: the number of ways is 9!

$$9! = 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 362,880$$

- In how many ways can 9 students set in 5 chairs

Solution: the number of ways is 9!

$$\frac{9!}{4!} = 9 \times 8 \times 7 \times 6 \times 5 = 15,120$$

Ordering same objects

Suppose you want to order a group of n objects where some of the objects are the same.

For instance, consider a group of letters consisting of four A's, 2 B's, and one C. How many ways can you order such a group?

Using the previous formula, you might conclude the following:

If we have n items, where $n_1, n_2, \dots, n_k$ are the same

Then the number of ways = $\frac{n!}{n_1!n_2!n_3!\dots n_k!}$, where $n_1 + n_2 + \dots + n_k = n$

Example

In how many ways can we order the following letters a, a, a, a, b, b, and c?

$$\begin{aligned} &= \frac{7!}{4! \times 2! \times 1!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4! \times 2! \times 1!} \\ &= \frac{7 \times 6 \times 5}{2} \\ &= 105 \end{aligned}$$

Permutations of n objects taken r at a time

- Suppose you want to choose some of the objects in a group and put them in order. Such an ordering is called a permutation of n objects taken r at a time.

$${}_n P_r = \frac{n!}{(n-r)!} \quad \text{Where } r \leq n$$

Example 5: Finding ${}_nP_r$

Suppose a doctor has a choice of 10 locations in which to establish her hospital. She decides to rank each location according to certain criteria such as price, parking facilities, etc. she only wanted to rank the top three. How many different ways can she rank them?

$${}_nP_r = {}_{10}P_3 = \frac{10!}{(10-3)!} = \frac{10!}{7!} = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

There are 720 possible ranking

Combinations

Suppose you want to buy three CD's from a selection of five CD's. There are 10 ways to make your selections

ABC, ABD, ABE, ACD, ACE, ADE, BCD, BCE, BDE, CDE.

In each selection, **order does NOT matter**. (ABC is the same set as BAC). The number of ways to choose r objects from n objects without regard to order is called the number of combinations of n objects taken r at a time.

Combination of Objects taken r at a time

- A combination is a selection of r objects from a group of n objects without regard to order and is denoted by ${}_n C_r$. The number of combinations of r objects selected from a group of n objects is:

$${}_n C_r = \frac{n!}{(n-r)!r!}$$

Example 7: Finding the number of combinations

A biology department plans to develop a new students activities the department want to choose four students committee out of 16. How many different combinations of four students can be selected from the 16 students? Because order is NOT important, there are:

$$\begin{aligned} {}^{16}C_4 &= \frac{16!}{(16-4)!4!} \\ &= \frac{16!}{12!4!} \\ &= \frac{16 \times 15 \times 14 \times 13}{4 \times 3 \times 2 \times 1} \\ &= \frac{43680}{24} = 1820 \end{aligned}$$

How to find probability

- If an experiment can result in any one of N different equally likely outcomes, and if exactly n of these outcomes correspond to event A , then the probability of event A is

$$P(A) = \frac{n}{N}$$

Applications – Example 8 Finding Probabilities

A word consists of one M, four I's, four S's, and two P's. If the letters are randomly arranged in order, what is the probability that the arrangement spells the word Mississippi?
Solution. There is one favorable outcome and there are

$$\begin{aligned} & \frac{11!}{1! \times 4! \times 4! \times 2!} \\ = & \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1 \times 2 \times 1} \\ & = \frac{1663200}{48} \\ & = 34650 \end{aligned}$$

There are 34,650 distinguishable permutations of the word Mississippi. So the probability that the arrangement spells the word Mississippi is:

Applications – Example 8 Finding Probabilities

There are 34,650 distinguishable permutations of the word Mississippi. So the probability that the arrangement spells the word Mississippi is:

$$P(\textit{Mississippi}) = \frac{1}{34,650} = .000029$$

Ex. 1: How many three-letter codes are there using letters A, B, C, and D if no letter can be repeated?

Note: The order does matter

$${}_4P_3 = \frac{4!}{1!} = 24$$

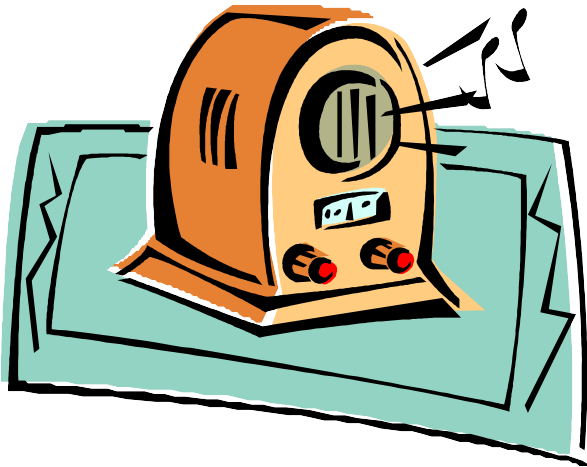
Ex. 2: How many committees of three can be selected from four people?

Use A, B, C, and D to represent the people

Note: Does the order matter?

$${}_4C_3 = \frac{4!}{3!1!} = 4$$

Ex. 3: How many ways can the 4 call letters of a radio station be arranged if the first letter must be W or K and no letters repeat? • $2 \cdot 25 \cdot 24 \cdot 23 = 27,600$



Ex. 5: How many five-card hands are possible from a standard deck of cards?



- ${}_{52}C_5 = 2,598,960$

Ex. 6: Given the digits 5, 3, 6, 7, 8, and 9 on cards, how many 3-digit numbers can be made if the first digit must be a prime number? (can digits be repeated?) (NO, Why?)

$$3 \cdot 5 \cdot 4 = 60$$

Ex. 8: Suppose there are 15 girls and 18 boys in a class. In how many ways can 2 girls and 2 boys be selected for a group project?

$${}_{15}C_2 \times {}_{18}C_2 = 16,065$$